

Aspect-based Sentiment Analysis on Hotel Reviews

UE24CS352A - Machine Learning Mini-Project | PES University | Section J
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1. Problem statement

A hotel review can praise a room while criticizing noise or price. A single overall rating hides these differences. This project learns to predict Positive, Neutral, or Negative sentiment and a 1-5 rating for individual hotel aspects, then presents those predictions in a local browser demonstration.

2. Dataset and label coverage

The experiment uses a seed-42 random sample of **20,000 reviews** from the accessible *jniimi/tripadvisor-review-rating* redistribution, whose source file contains 201,295 English reviews. Each input combines the review title and body. This dataset differs from the Wang dataset used in the supplied Yu paper [1, 2].

Targets	Available labels in this sample
Rooms, location, value, cleanliness, sleep quality	20,000 ratings for each aspect
Overall impression	20,000 ratings
Service and business service	No labels; prediction remains unavailable

Ratings 1-2 map to Negative, 3 to Neutral, and 4-5 to Positive. Missing ratings remain missing. Guest ratings provide review-level supervision; they do not identify the exact opinion sentence or establish that an aspect appears in the text.

3. Approach and evaluation design

We decode HTML entities, remove tags, lowercase text, and normalize spaces while retaining negation. Identical normalized reviews are removed before sampling. Reviewer groups define disjoint partitions: **12,007 training, 3,986 validation, and 4,007 test reviews**. Vocabulary and IDF weights use training text only. Hotel-independent evaluation is not established because verified hotel grouping is unavailable.

We compare word counts for unigrams/bigrams, counts for bigrams/trigrams, and TF-IDF for unigrams/bigrams, with a 40,000-feature cap. Candidate models include Multinomial Naive Bayes, balanced Linear SVM, a majority-class baseline, and Ridge regression for ratings. Each available target has an independent three-class sentiment model and a separate rating model. We select each model by validation macro-F1 and report held-out test performance [3].

4. Implementation

The Python project includes reproducible training, evaluation, a local HTTP demo, and automated checks. The corrected demo sends matching keyword excerpts to each aspect model, keeping unrelated opinions separate. Overall and optional unmentioned-aspect estimates use the complete review. Keyword extraction is heuristic. The measured scores below evaluate the original full-review baseline, not excerpt inference. Saved models, metrics, provenance, and split assignments support reproduction.

Results and conclusions

Full-review baseline on 4,007 held-out reviews. Selected models use validation macro-F1 with fixed hyperparameters. Excerpt inference needs separate accuracy evaluation.

5. Measured results

Target	Sentiment accuracy	Sentiment macro-F1	Exact rating accuracy
Overall	84.5%	0.713	64.4%
Rooms	80.9%	0.655	56.4%
Location	81.7%	0.513	62.6%
Value	77.1%	0.617	51.7%
Cleanliness	86.4%	0.624	60.6%
Sleep quality	79.5%	0.599	55.1%
Mean of five aspects	81.1%	0.602	57.3%

Sentiment models select TF-IDF SVM for rooms, value, and cleanliness, and count-based Naive Bayes for location and sleep quality. The five-aspect mean sentiment accuracy is **81.1%** and mean macro-F1 is **0.602**. Exact rating accuracy averages **57.3%**. These means weight aspects equally and exclude the overall target.

The majority baseline averages 78.7% sentiment accuracy. For location, the selected model scores 81.7% accuracy compared with 86.9% for always predicting the majority class. Its macro-F1 improves to 0.513 because minority classes matter. This illustrates why accuracy alone can overstate performance on positive-heavy reviews.

The supplied Yu study reports approximately 70-75% rating accuracy and 85-90% polarity accuracy [1]. These are historical reference findings. Different data, splits, label coverage, and model-selection procedures prevent a direct performance comparison.

6. Limitations and error analysis

A mixed review praising rooms, location, and value incorrectly gave sleep quality Positive, 5/5 despite "We did not sleep well in the uncomfortable bed." Routing the matching sleep excerpt to the same model changes this example to Negative, 1/5. Fifteen automated checks pass, including this regression. This example does not establish general accuracy. Models trained on full reviews may behave differently on excerpts. Keywords can miss implied aspects or combine different opinions. Sentiment and rating can disagree. Two aspects still lack labels.

7. Conclusion and future work

The project provides a working, reproducible classical machine learning baseline for five hotel aspects, with explicit handling of unavailable labels. The experiment supports aspect-level sentiment prediction, while minority-class performance and mixed-review errors limit reliability. Next steps are independently annotated aspect sentences, service/business-service labels, stronger aspect extraction, and evaluation with verified hotel groups.

References

- [1] Yangyang Yu. *Aspect-based Sentiment Analysis on Hotel Reviews*. Stanford CS229, Spring 2016. Supplied report and poster. cs229.stanford.edu/proj2016spr/report/032.pdf
- [2] Junichiro Niimi. *Hotel Review Dataset (English)*, 2024, redistribution associated with Li, Ott, and Cardie, EMNLP 2013. huggingface.co/datasets/jniimi/tripadvisor-review-rating
- [3] scikit-learn documentation. LinearSVC, F1 score, and common pitfalls. scikit-learn.org/stable/common_pitfalls.html